

Thomas Barroqueiro - Backend Engineer

✉ barroqt@gmail.com ☎ +33643021609 📄 in/thomas-barroqueiro 🐙 github.com/barroqt

SUMMARY

Backend software engineer with production experience in APIs, indexing pipelines, and blockchain integrations. Focused on business applications for supply chain & energy, plus payments, settlements, and real-world asset tokenization.

EXPERIENCE

Backend Engineer

Independent Work

December 2024 - Present

- **Semaphore:** Built an oracle that aggregates narrative and sentiment signals from public sources including X, Reddit, Google, and Telegram. Integrated x402 payments to run paid scraping services via Apify. Prototype placed 2nd at ETH Prague hackathon.
- **Outlier:** Refined LLM-generated coding responses through reinforcement learning

Backend developer

Momo networks

April 2023 - December 2024, Seoul, South Korea

- Built an automated indexing pipeline in Go to collect, normalize, and make searchable metadata for 400,000+ digital assets across Ethereum, Solana, and EVM-compatible networks.
- Used Meilisearch to improve metadata discoverability and query performance.
- Integrated external APIs and partner-facing embedding workflows for asset metadata distribution.
- Deployed and maintained services with PM2, monitoring, and production error tracking via Sentry.

Backend developer

Smart-Chain

November 2021 - February 2023, Paris, France

- Built NestJS backend services for digital asset platforms.
- Improved synchronization reliability and throughput through batch processing, parallel execution, RPC fallback strategies, and checkpointing.
- Worked on systems handling on-chain asset data and integrations with blockchain networks.
- Applied Factory and Decorator patterns where they helped keep the codebase modular, and used TDD.

Full-stack developer

Esens consulting

February 2020 - June 2021, Paris, France

- **Allianz trade:** Developed and integrated RESTful APIs using TypeScript and Node.js.
- **PMU:** Built full-stack features with vanilla JavaScript.
- **R&D:** experimented emerging AI and blockchain technologies

Full-stack developer

Palo-IT

April 2017 - February 2020, Singapore

- **DBS marketplace:** Developed a marketplace platform from early-stage team in a fast-growing environment.
- Built backend ETL services, authentication middleware, frontend-backend data interfaces.
- Built queue-based workflows with AWS SQS
- Implemented frontend and backend user behavior analytics to improve the overall user experience using adobe analytics.
- Solved scalability challenges using AWS lambdas and Serverless.
- Was reliable to onboard new members on the project.
- Technologies used: Node.js, AWS, Serverless, Redis, Nginx, adobe analytics, react.js, jenkins, git, jira.
- **NTUC Income:** Built a user Portal and insurance subscription platform for NTUC Income in react.js, TypeScript.
- **Ministry Of Education of Singapore:** Proof of Concept of an app facilitating the school registration process online in react.js. The project got approved and is currently implemented.

EDUCATION

Master in Software Engineering

同济大学 - Tongji University · Shanghai, China · 2017

Master of Information Technology

EPITECH - European Institute of Technology · Paris, France · 2017

CERTIFICATIONS

Alyra RS5000 Certification: Blockchain developer

Paris, France · 2023

- smart contract development in solidity, security, decentralized solutions and best practices.

COURSEWORK

C & C++ - Elementary programming - UNIX systems

EPITECH - European Institute of Technology · 2012 · C, C++

- **C**: linked lists, shell, distributed systems, chat server
- **C++**: openGL, virtual machine

INVOLVEMENT

Meet-up host

Gengoffee · December 2021 - Present

- Hosted international meet-ups in Paris and Seoul until 2024
- Currently hosting Korean-French language meet-ups and board game meet-ups twice a month in Paris

SKILLS

- **Backend**: Rust, Go, Node.js, TypeScript, JavaScript, NestJS, Express, REST, GraphQL
- **Cloud & Ops**: AWS (SQS, S3, EC2, API Gateway), Docker, Serverless, PM2, Sentry
- **Data**: SQL, PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Meilisearch, Prisma
- **Distributed systems & blockchain**: Ethereum, Solana, EVM, Solidity, Foundry, Hardhat, Hyperledger, IPFS
- **AI & LLM tools**: OpenCode, agentic coding workflows
- Spoken languages**: French, English, Mandarin Chinese, Korean, Portuguese, Spanish